# One map, then an exchange.

**A habit for scientists, a certificate for buyers, and a place where cell data converts.**

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## Vision

Every lab that images cells should know where its cells sit among everyone else's, and every dataset made from cells should carry that answer with it. Datum Line starts as a map, grows into a certificate, then an exchange: a place where cell data from any lab can be compared, trusted and shared by subscription. The arrow after the exchange is left open on purpose.

The habit comes first, the way Strava's segments place a run among everyone who ran the same stretch of road. Here the road is the cell line, and the field is every lab that ran it. Everything below serves one rule: **uploading must be effortless, and the answer must be worth looking at.**

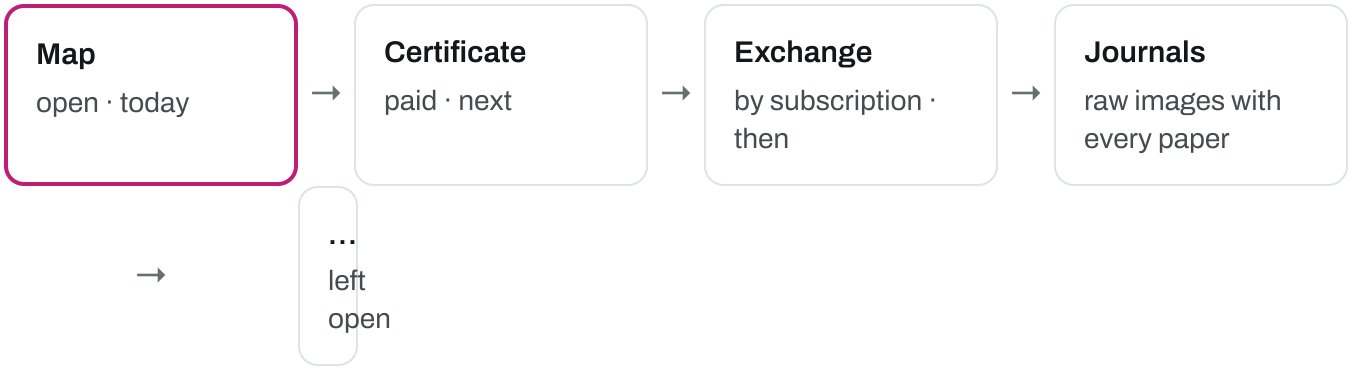
## 01 Why now

**Data is the new currency.** One vendor pays its experts more than $1.5M a day, about $85 an hour on average, at a $10B valuation. [10] Roche and Genentech paid Recursion a second $30M milestone for an accepted whole-genome cell map; $213M from the partnership to date. [11] A $1.8B virtual-cell data initiative, part new money and part existing funding; Meta, Google DeepMind and Isomorphic put in $300M together and get a year of exclusive access first. [12]

**Cell data does not convert.** EU-OPENSCREEN ran one Cell Painting protocol at 4 sites, with cells, serum and dye lots supplied centrally. Within a site, 84 to 94% of compounds replicate; across sites, 54 to 68% pair, and the authors call the profiles "not directly comparable between imaging sites" [13]. On the public JUMP data, a classifier names the lab from untreated wells alone at 99.9% against 9.1% chance. Whoever makes cell data comparable sets its exchange rate.

## 02 The path

Fig. 1 **A starting point, not an end state**



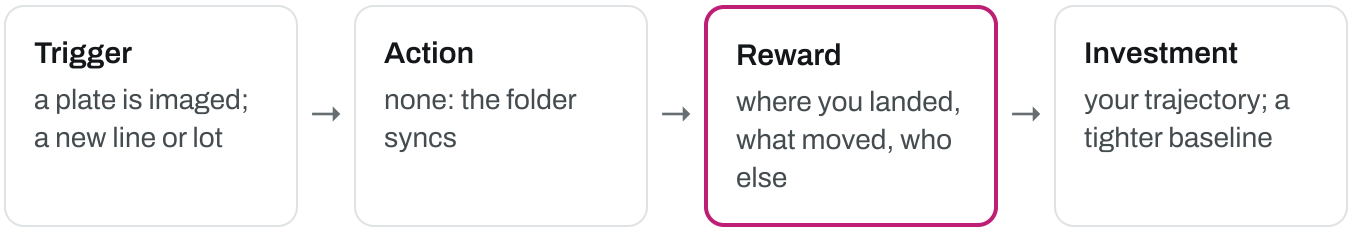

each stage pays for the next, and none is promised a date

| STAGE | WHAT IT IS | WHO PAYS | WHAT HAS TO BE TRUE | STATUS |
|---|---|---|---|---|
| **Map** | every lab's untreated wells placed on one map; the box, the code and the public reference | nobody: open, and never crippled | labs upload the controls they already image | today: 11 labs, 129 batches, 1,871 plates of public data |
| **Certificate** | where a dataset sits, who made it and what moved, attached to the data | companies that buy cell data, per dataset; groups that screen, per line per quarter | the map holds for the line | next: prices proposed on board A15 |
| **Exchange** | certified datasets, listed by the labs that made them | subscribers, for access to data from many labs, not ownership; each lab gets a share | buyers trust the certificate; labs choose to list | then |

| STAGE | WHAT IT IS | WHO PAYS | WHAT HAS TO BE TRUE | STATUS |
|---|---|---|---|---|
| … | what the map makes possible next: journals asking for raw control images with every paper, alerts when a lot drifts across labs, a graded benchmark, models trained on every lab's controls | | | open |

## 03 The habit loop

**Fig. 2** **Every upload makes the next answer worth opening**




↺ back to Trigger: the reference grows, the next reading is sharper

| STEP | IN DATUM LINE | DESIGN RULE |
|---|---|---|
| Trigger | the microscope finishes a plate; a new line or lot arrives | the trigger is the instrument, not an email |
| Action | nothing: the export folder syncs | any click is a leak; the reference starves on friction |
| Reward | where you landed, which features moved, who else moved | real uncertainty: nobody knows their position until they look |
| Investment | your history becomes a trajectory; your baseline tightens | leaving costs your own past, which is honest switching cost |

## 04 Six magic moments

Each moment turns an image the lab already has into something it could not see before, and each is ordered by when a new customer meets it.

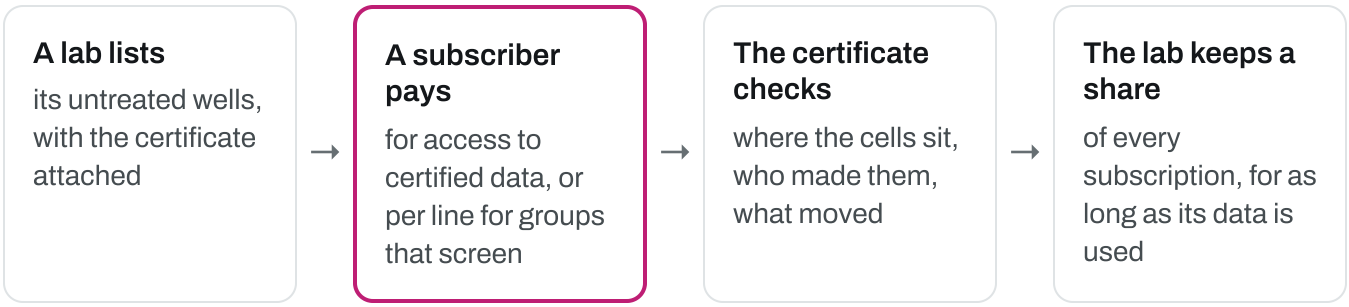
| # | MOMENT | WHAT THE LAB SEES | WHEN |
|---|---|---|---|
| 1 | **Your past, diagnosed** | old plates dragged in on day one appear as a path across the field map: "you left the envelope in March" | first ten minutes |
| 2 | **Show the cells, not the score** | when a plate moves, its nuclei next to their nearest neighbours from labs in distribution: same size, more texture | every flag |
| 3 | **The fix** | what the in-distribution labs do differently: seeding density, media or serum lot | after a flag |
| 4 | **A forecast** | "at your current drift, this line leaves the envelope in about five weeks" | after three readings |
| 5 | **Someone else's plate saves your screen** | a lot you use is drifting at other labs, before your own plates show it | once labs share lots |
| 6 | **The certificate as currency** | a verifiable link for a methods section, a CRO handoff, a grant report | when results leave the lab |

Moments 1 to 4 work for a single lab on day one, against the public reference. Moments 5 and 6 exist only once the network does, and no competitor can copy them without the labs.

## 05 The exchange

A certificate makes a dataset legible to someone who did not make it. The exchange is where it is shared: subscribers get access for a period, not ownership.

Fig. 3 **Every subscription pays the labs that made the data**




↺ every listing widens the map, and a wider map makes every certificate sharper

| WHAT SUBSCRIBERS GET | WHY IT IS WORTH PAYING FOR |
|---|---|
| **Certified datasets** | they convert: on the map, agreement between labs on the same drugs goes from 0.75 to 0.78 (board A14) |
| **Early access** | a window before data goes public, which consortia already sell: a year, at the virtual-cell initiative [12] |
| **The benchmark** | AI labs and model builders graded on planted failures, with a cell-count baseline on every task (board A16) |

**What never gets sold.** A lab's treated data, its results, its name without consent, a ranking of instruments, or a verdict for hire. The certificate can be bought; the answer cannot. Prices and shares are proposals (boards A15 and A22) until the first partner labs and the first buyer set them.

## 06 Trust, without the token

The open-science half of the product, and the reason a lab would share.

| PROMISE | HOW | STATUS |
|---|---|---|
| **Control wells only** | nothing else leaves a lab: no compound, no target, no result; the frame is applied on the lab's side | today |

| PROMISE | HOW | STATUS |
|---|---|---|
| **Every lab leaves a fingerprint** | 99.9% of untreated wells name their own lab on plates the model never saw, so a dataset claimed from one lab and made in another is caught | today |
| **Signed and timestamped** | every certificate signed, its fingerprint written to a public log: anyone can check it, nobody can backdate it; a blockchain is one such log, not a requirement | proposed |
| **The lab keeps its data** | listing is a choice and can be withdrawn; a share of every subscription goes to the lab that made the data | proposed |
| **Known buyers** | buyers verified at sign-up; unusual access flagged; every access in a log the members can read | proposed |
| **Low risk by design** | images of cell lines: no human genetic data and no pathogen data; the map holds untreated wells only | today |

For a decentralized-science audience this is the point: open data, verifiable records and contributor upside, without a token to speculate on and without betting on anyone's results.

## 07 The game

Every batch a lab sends is ranked against every batch the field has seen, on its untreated wells alone: how typical the cells are, how alike the plates are, how steady the lab is from batch to batch, and cells per well against the field. Nothing in the score touches a treated well or a result, so the only way to climb is better culture and imaging practice. On the public data, lab 5 holds the field's longest streak, 33 batches (results/e14). A lab sees its rank, its personal best, its streak and the lab to beat.

| DO | DON'T | WHY |
|---|---|---|
| rank practice, never results | rank a lab's results or its drugs | the only way up is better culture and imaging |
| labs are numbers until they choose a name | publish a lab's name or rank without consent | shame makes labs hide their worst plates, which teach the field most |

| DO | DON'T | WHY |
|---|---|---|
| credit for uploading flagged plates | penalise a batch for sitting off the map | a failure uploaded is worth more than a success |
| explain every flag with cells and a likely fix | show a bare red dot | a verdict without a reason reads as judgement |
| let the instrument trigger uploads | send nudges to protect a streak | the reward is the answer, not the visit |

## 08 Metrics

North star: the share of a lab's plates that upload with no human action. If it is high, the loop turns; if people have to remember, nothing else matters.

| METRIC | WHAT IT TELLS YOU |
|---|---|
| automatic upload share | the habit exists without effort |
| flagged plates uploaded, not deleted | labs trust the product with bad news |
| labs per line, lines per lab | the map is widening |
| datasets listed with a certificate | labs trust the exchange with their data |
| renewing subscribers | access is worth paying for, again |
| share of each subscription paid to the lab | contributors are earning, so more will list |
| certificates cited outside the lab | the certificate has become currency |

## 09 What could go wrong

| RISK | WHY IT IS REAL | ANSWER |
|---|---|---|
| cold start | no map without labs, no labs without a map | seed from JUMP's public controls; U2OS first, any other line from its first batch |
| labs will not list | data is a lab's edge, and pharma data is intellectual property | the certificate is free and useful alone; listing is optional and pays the lab |
| public data must stay free | NIH-funded labs must share the data behind their findings | charge for certification, comparability and compute, never for access to data a lab owes the public |
| buyers want designed data at scale | the deals above buy factories and consortia, not leftovers | certify factory and consortium data too; the benchmark grades their models |
| a platform builds its own map | phenomics companies hold the most images | nobody can certify their own drift; the reference has to be neutral and open |

## References


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